

Content

Learning objectives and content

Objectives

At the end of the course the student (in a team) has experienced and is able to

Technical Learning Objectives:

- select and implement distributed control algorithms that run on a set of autonomous robots;
- select and integrate new sensors and actuators in an existing control system;
- design and implement a robust wireless communication protocol;
- report on the system design and its individual components, with an integrated P&PD Skills.

Systems Thinking:

- explain the relevance of given design requirements/preferences/constraints (RPC's) and can articulate the technical design assignment;
- analyze the system RPC's, decompose it into a set of smaller sub-systems, and derive RPC's for the sub-systems and specify their mutual interaction;
- explain the role of (sub-)system boundaries, derive interface requirements between the sub-systems based on the given system requirements, derive sub-system requirements from the given system requirements;
- assess system implications of different implementations on sub-system level

System Design (Systems Engineering models):

- apply the iterative V-model or an alternative SE model (e.g., 1. requirements analysis, 2. design phase, 3. validation/verification/measurements);
- identify multiple possible solutions and select a preferred one that is based on their implications from the view of the overall system and that is compliant with the RPCs;
- can design, based on a well-founded design strategy, and build the proposed solution;
- can integrate all sub-modules into a complete working system;
- can evaluate the solution's performance, and draw relevant conclusions with a focus on the derived RPCs and performance evaluations.

Content

During this CBL exercise, the students work in a team to design and build a system that is used to explore an environment (assuming the surface of planet Venus). Firstly, students need to consider an effective team set-up and role division. The teams are asked to develop robots equipped with a variety of sensors and actuators which explore the surface of the planet, find rock samples, pick them, and take them to a central station (lab). These robots need to connect to each other and cooperate to complete the requested challenging tasks. There are limitations with respect to environment, operation time, battery, and size of the robot pack that need to be considered by the teams when they are designing and implementing their system. System engineering and reporting aspects are important parts of the assignment.

Learning Type

Challenge based

2026

Engineering challenge for Venus (5EID0)

Practical information

Course module

5EID0

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Deepening

Language of instruction

English

Follow-up subjects

Autonomous Vehicles (5AID0) OGO, Electronic systems engineering (5XIC0), BEP graduation project

Offered by

Electrical Engineering - Electronic Systems

Remark

Pay attention! Maximum number of participants is 360. See information below about the waiting list and who has priority:

General course information

If this course is not part of your curriculum as a core course, you will be placed on the waiting list until the registration deadline has passed. After the registration deadline, we will check which of the students on the waiting list can be added to the course. The maximum capacity of the course is 360 students. Which means that students from the waiting list can only be added until the maximum capacity is reached. You will receive an email from the CSA of EE if you are not able to take the course.

If you have permission of the examination committee to take this course as part of your program, please contact ee.csa@tue.nl.

Please make sure to register for a back-up course in case you cannot take the course due to the capacity limitation.

Course enrolment

Waiting list

For this course a waiting list applies. The procedure and criteria for placement can be found in the course catalogue.

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4

Timeslot(s)

D - We 5-8, Tu 9-10, Fr 1-4

Enrolment period

15 November 2026 until 21 March 2027 23:59

Unenrolment

until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge

Basic knowledge on electrical and electronic circuits, programming, sensors and actuators

Previous knowledge can be gained by

Programming and engineering challenge (5EWC0) and Circuits (5EPC0)

Instructional modes

Instructional modes

- DBL-group meeting

Remark

- Lecture

Remark

Tests

Tests

- Assignment

Interim examination

Test weight

15

Minimum grade

Assessment

Since assessment is based on the team project, there will be no resit.

Remark

Since assessment is based on the team project, there will be no resit.

- Demonstrator / Demonstration

Interim examination

Test weight

15

Minimum grade

Assessment

Since assessment is based on the team project, there will be no resit.

Remark

Since assessment is based on the team project, there will be no resit.

- Report

Final examination

Test weight

70

Minimum grade

Assessment

Since assessment is based on the team project, there will be no resit.

Remark

Since assessment is based on the team project, there will be no resit.

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 4

Enrolment period

15 November 2026 until 6 June 2027 23:59

Demonstrator / Demonstration

- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Report

- Block TJAAR
Enrolment period
15 June 2026 until 31 August 2027 23:59

Lecturer(s)

Co-lecturer

- dr. M. Nabi , MSc
- dr.ir. S. Stuijk

Contactperson for the course

- dr.ing. L.T. Maile

Responsible lecturer

- dr.ing. L.T. Maile

Subject matter expert

- dr.ing. L.T. Maile

Materials

Required material

- Hardware setup (robots' platform, communication modules, sensors, electronic components) are provided to the student teams

Do you want to know more?

dr.ing. L.T. Maile can tell you more about this course.

Send an e-mail to l.t.maile@tue.nl